Test call Between MS Teams and freePBX

Prerequisites Checklist

Before making any call, verify every item below is green.

Pre-Check 1 — Verify All SIP Options Are Healthy

On FreePBX:

bash

sudo asterisk -rx "pjsip show contacts"

Expected output:

AudioCodes-SBC/sip:172.31.9.171:5067 Avail ~7ms

101/sip:101@YOUR-PC-IP:5060 Avail ~Xms

On AudioCodes → Monitor → VoIP Status → Proxy Sets Status:

IP-PBX | ONLINE ✅

ITSP | ONLINE ✅

On Teams Admin Center → Direct Routing:

mylab-sbc.ddns.net

TLS Status: Active ✅

SIP Options: Active ✅

Pre-Check 2 — Verify Number Mapping

User	Teams Number	FreePBX Extension	Device
Anoop	+15550101	—	Teams on laptop
Test	—	101	MicroSIP on laptop

Pre-Check 3 — Verify FreePBX Inbound Routes

Go to **FreePBX → Connectivity → Inbound Routes**:

You need a route that maps the DID coming from Teams/SBC to MicroSIP extension 101.

Field	Value
Description	Teams-to-Ext101
DID Number	15550101
Destination	Extensions → 101

Also add:

Field	Value
DID Number	+15550101
Destination	Extensions → 101

Click **Submit → Apply Config**

The + prefix variant is needed because Teams sends E.164 format.

Pre-Check 4 — Verify FreePBX Outbound Route

Go to **Connectivity → Outbound Routes → To-Teams-via-SBC**:

Dial Patterns tab — must have ALL of these:

Pattern	Description
+NXXXXXXXXXXXX	E.164 with + prefix 11 digits
+X.	Any E.164 with +
1NXXXXXXXXXX	11 digit with country code
NXXXXXXXXXX	10 digit US number

Trunk Sequence: AudioCodes-SBC

Click **Submit → Apply Config**

Pre-Check 5 — Verify AudioCodes IP-to-IP Routing

Go to **AudioCodes → Signaling & Media → SBC → Routing → IP-to-IP Routing**:

Index	Source IP Group	Request Type	Destination IP Group
10	IP-PBX	INVITE	ITSP

Index	Source IP Group	Request Type	Destination IP Group
20	ITSP	INVITE	IP-PBX

Both rules must exist with **INVITE** only — not All.

Pre-Check 6 — Verify Number Manipulation (Critical)

Teams sends numbers as +15550101 but FreePBX extension is just 101. You need manipulation rules in AudioCodes.

Go to **Signaling & Media → SBC → Manipulation → Inbound Manipulations** (ITSP → IP-PBX direction):

Field	Value
Name	Teams-DID-to-Extension
Source IP Group	ITSP
Destination IP Group	IP-PBX
Manipulated URI	Destination
Remove From Left	2 (removes +1 leaving 5550101)

Note: Check what DID FreePBX receives — you may need to strip more or less. We'll verify during testing.

Step-by-Step Testing

Test 1 — MicroSIP Extension 101 → Teams User Anoop (+15550101)

This tests the **outbound call path**: MicroSIP → FreePBX → SBC → Teams.

Step 1.1 — Enable live logging on FreePBX:

bash

```
sudo asterisk -rx "core set verbose 5"

sudo tail -f /var/log/asterisk/full | grep -E
"INVITE|DIAL|Dial|route|trunk|101|15550101|ERROR|WARNING"
```

Step 1.2 — Enable SBC Message Log: Go to **AudioCodes → Troubleshoot → Message Log → Start Logging**

Step 1.3 — Make the call: In MicroSIP, dial: +15550101

Step 1.4 — Expected flow:

MicroSIP → INVITE → FreePBX:5060

FreePBX matches outbound route "To-Teams-via-SBC"

FreePBX → INVITE → AudioCodes SBC:5067

SBC manipulates number → sends to Teams

Teams → rings Anoop's Teams client on laptop

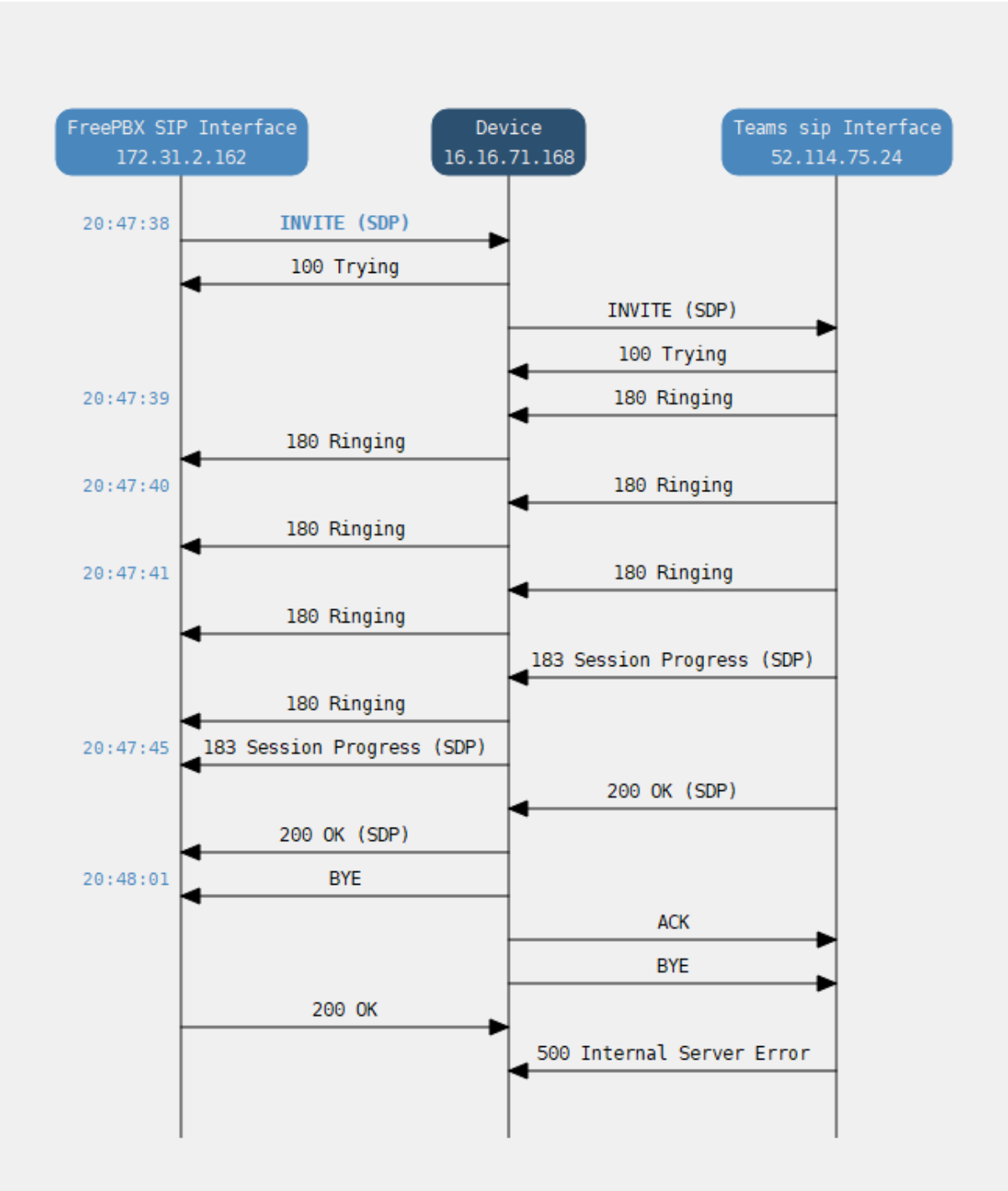
Anoop answers → two-way audio established

Step 1.5 — What to watch in Asterisk log:

-- Called PJSIP/+15550101@AudioCodes-SBC

-- PJSIP/AudioCodes-SBC is ringing

-- PJSIP/AudioCodes-SBC answered



Test 2 — Teams User Anoop (+15550101) → MicroSIP Extension 101

This tests the **inbound call path**: Teams → SBC → FreePBX → MicroSIP.

Step 2.1 — Enable live logging:

bash

sudo tail -f /var/log/asterisk/full | grep -E "INVITE|Executing|Dial|101|15550101|from-trunk|ERROR"

Step 2.2 — Make the call from Teams: In Teams client (Anoop), open dialpad → dial +15550101

Since Anoop IS +15550101, Teams may not allow calling yourself. If so, use a second Teams user or test from Teams Admin Center.

Alternative — dial MicroSIP from Teams Admin Center: Go to **Teams Admin Center → Users → Anoop → Test call**

Step 2.3 — Expected flow:

Teams → INVITE → SBC:5061 (TLS)

SBC classifies as ITSP → routes to IP-PBX

SBC → INVITE → FreePBX:5060

FreePBX matches inbound route DID 15550101 → Extension 101

Extension 101 → MicroSIP rings on laptop

Answer → two-way audio

Step 2.4 — What to watch in Asterisk log:

-- Executing Dial(PJSIP/101)

-- PJSIP/101@0.0.0.0-00000001 is ringing

Common Errors and Fixes

Error 1 — "404 Not Found" when calling from MicroSIP to Teams

Cause: Outbound route dial pattern doesn't match what MicroSIP is dialing, or trunk not sending correctly to SBC.

Fix:

bash

Check what number FreePBX is trying to dial

sudo tail -f /var/log/asterisk/full | grep "Dial\|INVITE"

In FreePBX → Outbound Route → add more permissive pattern: +X.

Also check SBC Message Log — the INVITE destination number must be in E.164.

Error 2 — "503 Service Unavailable" from SBC

Cause: SBC can't reach Teams or IP-PBX proxy is down.

Fix:

bash

```
# Check SBC proxy status
sudo asterisk -rx "pjsip show contacts"
# AudioCodes-SBC must show Avail
```

On AudioCodes → Monitor → check both Proxy Sets are ONLINE.

Error 3 — Teams Rings but "480 Temporarily Unavailable" on inbound

Cause: Inbound route DID doesn't match what SBC sends to FreePBX.

Debug:

bash

```
sudo tail -f /var/log/asterisk/full | grep "DID\|extension\|INVITE\|from-trunk"
```

Check what DID arrives at FreePBX:

bash

```
sudo asterisk -rx "core set verbose 10"
# Then trigger a call from Teams and look for:
# "Executing" lines showing the dialed extension
```

Fix options:

Option A — Add both DID formats to inbound route:

DID: 15550101

DID: +15550101

DID: 5550101

Option B — Add a "catch-all" inbound route:

DID Number leave blank

Destination Extension 101

Error 4 — Call connects but NO AUDIO (one-way or no audio)

This is the most common NAT issue with cloud deployments.

Cause A — FreePBX sending private IP in SDP:

bash

```
# Check what IP FreePBX advertises in SDP
sudo asterisk -rx "pjsip set logger on"
sudo tail -f /var/log/asterisk/full | grep "c=IN"
```

If it shows c=IN IP4 172.31.4.212 (private IP) — NAT fix needed.

Fix: Go to **FreePBX → Admin → SIP Settings → Chan PJSIP**:

External Address: 13.51.250.68

Local Networks: 172.31.0.0/16

Also on each extension → Advanced:

Direct Media: No

Force rport: Yes

Rewrite Contact: Yes

RTP Symmetric: Yes

Cause B — RTP ports blocked:

bash

```
# On FreePBX check RTP port range
sudo asterisk -rx "rtp show settings"
```

Make sure AWS Security Group has UDP 10000-20000 open.

Cause C — SBC not transcoding RTP correctly:

On AudioCodes → IP Profile for IP-PBX:

Media Security: Disable (FreePBX doesn't use SRTP by default)

RTP Mode: Regular

Error 5 — MicroSIP shows "Request Timeout" still

Cause: Transport not set on extension or UFW blocking.

Fix:

bash

```
# On FreePBX
sudo ufw allow 5060/udp
sudo ufw allow 10000:20000/udp
sudo ufw reload

# Reload Asterisk
sudo fwconsole reload
sudo asterisk -rx "module reload res_pjsip.so"
```

In FreePBX → Extension 101 → Advanced:

Transport: 0.0.0.0-udp

Direct Media: No

Error 6 — "488 Not Acceptable Here" — Codec mismatch

Cause: Teams uses G.711 (ulaw/alaw) and SILK. If FreePBX or SBC restricts codecs, call fails.

Fix — FreePBX: Go to **Admin → SIP Settings → Codec Settings**: Enable: ulaw, alaw, g722 (in this order)

Fix — Extension 101:

allow: ulaw,alaw,g722

Fix — AudioCodes IP Profile for ITSP:

Allowed Coders: PCMU, PCMA, G.722

Error 7 — Teams shows "We couldn't complete this call"

Cause: Voice route not matching the number format or SBC FQDN mismatch.

Fix — PowerShell:

powershell

```
Connect-MicrosoftTeams -UseDeviceAuthentication

# Check voice routes
Get-CsOnlineVoiceRoute | Format-List Identity, NumberPattern, OnlinePstnGatewayList

# Check user assignment
Get-CsPhoneNumberAssignment -Identity "AnoopR@Voicenet966.onmicrosoft.com"

# If route missing, recreate
New-CsOnlineVoiceRoute -Identity "Route-Ext101" `
  -NumberPattern "^\+15550101$" `
  -OnlinePstnGatewayList "mylab-sbc.ddns.net" `
  -Priority 1 `
  -OnlinePstnUsages "LabUsage"
```

Quick Reference Debug Commands

bash

```
# Live call trace - run BEFORE making test call
sudo asterisk -rx "pjsip set logger on"
sudo asterisk -rx "core set verbose 5"
sudo tail -f /var/log/asterisk/full

# Check all registered contacts
sudo asterisk -rx "pjsip show contacts"

# Check active channels during call
```

```
sudo asterisk -rx "core show channels"

# Check codec negotiation
sudo asterisk -rx "core show channel PJSIP/101-xxxxxxxx"

# Check RTP stats after call
sudo asterisk -rx "rtp show settings"
```

Summary of Test Call Matrix

From	To	Dial	Expected Result
MicroSIP ext 101	Teams Anoop	+15550101	Teams rings
Teams Anoop	MicroSIP ext 101	+15550101	MicroSIP rings
MicroSIP ext 101	MicroSIP ext 102	102	Internal ring